

How do we know if we have enough data? From the probability distribution—a distribution does tell you whether you have enough data to “build confidence” about what you are inferring. If it is a Gaussian bell curve, then a few points will suffice (the law of large numbers once again). And how do you know if the distribution is Gaussian? Well, from the data. So we need the data to tell us what the probability distribution is, and a probability distribution to tell us how much data we need. This causes a severe regress argument.

This regress does not occur if you *assume beforehand* that the distribution is Gaussian. It happens that, for some reason, the Gaussian yields its properties rather easily. Extremistan distributions do not do so. So selecting the Gaussian while invoking some general law appears to be convenient. The Gaussian is used as a default distribution for that very reason. As I keep repeating, assuming its application beforehand may work with a small number of fields such as crime statistics, mortality rates, matters from Mediocristan. But not for historical data of unknown attributes and not for matters from Extremistan.

Now, why aren't statisticians who work with historical data aware of this problem? First, they do not like to hear that their entire business has been canceled by the problem of induction. Second, they are not confronted with the results of their predictions in rigorous ways. As we saw with the Makridakis competition, they are grounded in the narrative fallacy, and they do not want to hear it.

ONCE AGAIN, BEWARE THE FORECASTERS

Let me take the problem one step higher up. As I mentioned earlier, plenty of fashionable models attempt to explain the genesis of Extremistan. In fact, they are grouped into two broad classes, but there are occasionally more approaches. The first class includes the simple rich-get-richer (or big-get-bigger) style model that is used to explain the lumping of people around cities, the market domination of Microsoft and VHS (instead of Apple and Betamax), the dynamics of academic reputations, etc. The second class concerns what are generally called “percolation models,” which address not the behavior of the individual, but rather the terrain in which he operates. When you pour water on a porous surface, the structure of that surface matters more than does the liquid. When a grain of sand hits a pile of other grains of sand, how the terrain is organized is what determines whether there will be an avalanche.

Most models, of course, attempt to be precisely predictive, not just descriptive; I find this infuriating. They are nice tools for illustrating the